

Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step
$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$
$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + Q(k-1)$$

S2 – A Posteriori Update / Measurement Update
$$P_m(k) = \left(P_p^{-1}(k) + H^\top(k)R^{-1}(k)H(k)\right)^{-1}$$
$$\hat{x}_m(k) = \hat{x}_p(k) + \underbrace{P_m(k)H^\top(k)R^{-1}(k)}_{\text{correctionfactor}} \underbrace{(\bar{z}(k) - H(k)\hat{x}_p(k))}_{\text{predictionerror}}$$

Alternative Equations – all matrices/vectors at time k
S2 – Kalman Filter Gain:
$$K = P_p H^\top \left(H P_p H^\top + R \right)^{-1}$$
$$\hat{x}_m = \hat{x}_p + K \left(\bar{z} - H \hat{x}_p \right)$$
$$P_m = \underbrace{(\mathbb{I} - KH) P_p}_{\text{cheaper}} = \underbrace{(\mathbb{I} - KH) P_p (\mathbb{I} - KH)^\top + K R K^\top}_{\text{fewer numerical errors}}$$

Equivalent to RLS $\rightsquigarrow$ among linear, unbiased estimators KF minimizes MSE

KF Remarks
Known Data $A(k), H(k), Q(k), R(k) \forall k$ as well as $P_0 \rightsquigarrow$ KF matrices $P_p(k), P_m(k), K(k)$ can be computed offline

$$\hat{x}_p(k) = A(k-1)\hat{x}_m(k-1) + u(k-1)$$
$$\hat{x}_m(k) = \underbrace{(\mathbb{I} - KH)A(k-1)}_{\hat{A}(k)} \hat{x}_m(k-1) + \underbrace{(\mathbb{I} - KH)u(k-1) + K\bar{z}(k)}_{\hat{B}(k)}$$

KF not guaranteed to be optimal – non-linear estimators may do better

4.1.1 Asymptotic Properties of KF
Model assume **TI System, stationary Distributions** (A, H, Q, R) constants)
Change of Variance

$$P_p(k+1) = \left(A P_p(k) A^\top \right) + Q$$
$$- \left(A P_p(k) H^\top \right) \left(H P_p(k) H^\top + R \right)^{-1} \left(H P_p(k) A^\top \right)$$

Scalar Case $P_p(k+1) = \frac{a^2 r P_p(k)}{h^2 P_p(k) + r} + q =: f(P_p(k))$

Summary *Variance Converges to unique steady-state sol'n provided either $|a| < 1$ or if $|a| \geq 1 \Rightarrow h \neq 0$ and $q > 0$*
If slope for small P_∞ is $> 1 \Rightarrow$ unstable ($|a| > 1$)

4.1.2 Detectability / Stabilizability
Detectability Conditions
Pair (A, H) detectable
 $\Leftrightarrow$ For deterministic LTI Sys. $(x(k)=Ax(k-1), z(k)=Hx(k))$
 $\lim_{k \rightarrow \infty} z(k) = 0 \Rightarrow \lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$
 $\Leftrightarrow \left[A^{-\lambda} \mathbb{I} \right]$ is full rank for all $\lambda \in \mathbb{C}$ with $|\lambda| \geq 1$ (PBH-Test)
 $\Leftrightarrow$ Eigenvalues of $A - LH$ (or $(\mathbb{I} - LH)A$) can be placed within unit circle by suitable choice of $L \in \mathbb{R}^{n \times m}$

Stabilizability Conditions
Pair (A, B) stabilizable
 $\Leftrightarrow$ For deterministic LTI Sys. $(x(k)=Ax(k-1)+Bu(k-1))$
 $\exists u(0:k-1)$ s.t. $\lim_{k \rightarrow \infty} x(k) = 0 \quad \forall x_0 \in \mathbb{R}^n$
 $\Leftrightarrow [A - \lambda \mathbb{I} \ B]$ full rank $\forall \lambda \in \mathbb{C}$ with $|\lambda| \geq 1$ (PBH-Test)
 $\Leftrightarrow$ Eigenvalues of $A - BK$ (or $(\mathbb{I} - BK)A$) can be placed within unit circle by suitable choice of $K \in \mathbb{R}^{m \times n}$

NOTE Stabilizability is dual of detectability, (A, B) stabilizable
Observability / Reachability
if $(A^\top, B^\top)$ detectable
Observability Matrix implies detectability $\mathcal{O} = \begin{bmatrix} H \\ H A^{n-1} \end{bmatrix}$
Reachability Matrix implies stabilizability $\mathcal{R} = \begin{bmatrix} A & A^{n-1} B \end{bmatrix}$

4.1.3 Steady-State KF
DARE – Discrete Algebraic Riccati Equation (P_∞ is P_p)
$$P_\infty = A P_\infty A^\top + Q - A P_\infty H^\top (H P_\infty H^\top + R)^{-1} H P_\infty A^\top$$

Steady-State KF Gain $K_\infty = P_\infty H^\top (H P_\infty H^\top + R)^{-1}$
Filter Equations Linear Time-Invariant System
$$\hat{x}(k) = \underbrace{(\mathbb{I} - K_\infty H)A}_{\hat{A}} \hat{x}(k-1) + \underbrace{(\mathbb{I} - K_\infty H)u(k-1) + K_\infty \bar{z}(k)}_{\hat{B}}$$

Error Dynamics (also holds for non ss-KF) $e(k) = x(k) - x_f(k)$
$$e(k) = (\mathbb{I} - K_\infty H)A e(k-1) + (\mathbb{I} - K_\infty H)v(k-1) - K_\infty w(k)$$
$$\mathbb{E}\{e(k)\} = (\mathbb{I} - K_\infty H)A \mathbb{E}\{e(k-1)\}$$

- $(\mathbb{I} - K_\infty H)A$ must be stable (all $|\lambda| < 1$) for e not to diverge
- $\mathbb{E}\{e(k)\} \xrightarrow{k \rightarrow \infty} 0$ iff $(\mathbb{I} - K_\infty H)A$ stable

Theorem for $R > 0, Q \geq 0, G$ s.t. $Q = GG^\top$
 $\Leftrightarrow (A, H)$ is detectable, (A, G) is stabilizable
 $\Leftrightarrow$ DARE has unique pos. semidef. sol. $P_\infty \geq 0$, resulting $(\mathbb{I} - K_\infty H)A$ is stable and
$$\lim_{k \rightarrow \infty} P_p(k) = P_\infty \quad \forall P_p(1) \geq 0$$

(A,H) Detectable: can observe all unstable modes
(A,G) Stabilizable: noise excites unstable modes

Note if $Q > 0 \Rightarrow (A, G)$ always stabilizable
If $Q = 0$ multiple valid solutions can be possible

4.2 Extended Kalman Filter
Extension of KF to non-linear systems:
$$x(k) = q_{k-1}(x(k-1), v(k-1))$$
$$z(k) = h_k(x(k), w(k))$$

$x(0), v(\cdot), w(\cdot)$ mutually ind.	$x(0) \sim \mathcal{N}(x_0, P_0)$
$v \sim \mathcal{N}(0, Q(k))$	$w \sim \mathcal{N}(0, R(k))$

EKF Equations
Initialization: $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step
$$\hat{x}_p(k) = q_{k-1}(\hat{x}_m(k-1), 0)$$
$$P_p(k) = A(k-1)P_m(k-1)A^\top(k-1) + L(k-1)Q(k-1)L^\top(k-1)$$
$$A(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial x}, \quad L(k-1) = \frac{\partial q_{k-1}(\hat{x}_m(k-1), 0)}{\partial v}$$

S2 – A Posteriori Update / Measurement Update (at k)
$$K = P_p H^\top \left(H P_p H^\top + M R M^\top \right)^{-1}$$
$$\hat{x}_m = \hat{x}_p + K \left(\bar{z} - h_k(\hat{x}_p(k), 0) \right)$$
$$P_m = (\mathbb{I} - K(k)H(k)) P_p(k)$$
$$H(k) := \frac{\partial h_k(\hat{x}_p(k), 0)}{\partial x}, \quad M(k) := \frac{\partial h_k(\hat{x}_p(k), 0)}{\partial w}$$

Intuition
Process Update – predict the mean state estimate fwd using NL process model and update the var. according to lin. eqns
Measurement Update – correct for mismatch between $\bar{z}(k)$ and prediction $h_k(\hat{x}_p(k), 0)$, and correct var. according to lin. eqn

Cautions

- Process & measurement noise **assumed** 0-mean
- A, L, H, M lin. about current $\hat{x}_m \rightsquigarrow$ **cannot** compute offline
- Prior update only accurate if $\mathbb{E}\{\cdot\} \cdot q_{k-1}(\cdot)$ commute

EKF Variables DO NOT capture true conditional mean & var.
4.2.1 Hybrid EKF
System Process – non-lin, CT, **Measurement** – non-lin, DT
$$\dot{x}(k) = q(x(t), v(t), t) \quad x[k] := x(kT)$$
$$z[k] = h_k(x[k], w[k]) \quad \mathbb{E}\{w[k]\} = 0, \text{Var}\{w[k]\} = R$$

CT White Noise $\delta(t)$ = Dirac pulse
$$\mathbb{E}\{v(t)\} = 0 \text{ and } \mathbb{E}\{v(t)v^\top(t+\tau)\} = Q_c \delta(\tau)$$

True CT White noise does not exist $\rightarrow$ Would require infinite power
Hybrid EKF Equations
Initialization $\hat{x}_m(0) = x_0, \quad P_m(0) = P_0$
S1 – Prior Update / Prediction Step (Solve ODEs)
Solve, for $(k-1)T \leq t \leq kT$ and $\hat{x}((k-1)T) = \hat{x}_m[k-1]$
$$\dot{\hat{x}}(t) = q(\hat{x}(t), 0, t) \Rightarrow \hat{x}_p[k] := \hat{x}(kT)$$

Solve, for $(k-1)T \leq t \leq kT$ and $P((k-1)T) = P_m[k-1]$
$$\dot{P}(t) = AP + PA^\top + LQ_c L^\top \Rightarrow P[k] := P(kT)$$

with
$$A(t) := \frac{\partial q}{\partial x}(\hat{x}(t), 0, t), \quad L(t) := \frac{\partial q}{\partial v}(\hat{x}(t), 0, t)$$

S2 – A Posteriori Update / Measurement Update
Measurement model still DT – S2 identical to standard EKF

Caution

- Process & measurement noise **assumed** 0-mean
- ODEs typically solved numerically: Accuracy $\uparrow$ = Comp. Cost $\uparrow$

5 Particle Filter
5.1 Monte Carlo Sampling
Samples N DRV's y^n of samples of $y \in \mathcal{Y} = \{1, \dots, \bar{Y}\} \sim p_y$
Helper Variables
$$s_i^n = \delta(i - y^n) = \begin{cases} 1 & \text{if } y^n = i \\ 0 & \text{otherwise} \end{cases} \quad \left| \begin{array}{l} i = 1, \dots, \bar{Y} \\ n = 1, \dots, N \end{array} \right.$$

Average – s_i converges to $p_y(i)$ by Law of Large Numbers

$$\frac{1}{N} \sum_{n=1}^N s_i^n \xrightarrow{N \rightarrow \infty} \mathbb{E}\{s_i^n\} = p_y(i) \quad p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \bar{s}_i^n$$

Approximation
For $y \in \mathcal{Y}$, it holds $p_y(i) \approx \frac{1}{N} \sum_{n=1}^N \delta(i - \bar{y}^n), \quad i \in \mathcal{Y}$
For $x = g(y)$, with $x \in \mathcal{X} := g(\mathcal{Y})$, it holds

$$p_x(j) \approx \frac{1}{N} \sum_{n=1}^N \delta(j - g(\bar{y}^n)), \quad j \in g(\mathcal{Y})$$

Monte Carlo Sampling CRV
Helper Variables
$$s_a^n = \int_a^{a+\Delta y} \delta(i - y^n) d\xi = \begin{cases} 1 & \text{if } a \leq y^n < a + \Delta y \\ 0 & \text{otherwise} \end{cases}$$

Average – s_a converges to $p_y(\xi)$ by Law of Large Numbers
$$\int_a^{a+\Delta y} \frac{1}{N} \sum_{n=1}^N \delta(\xi - y^n) d\xi \xrightarrow{N \rightarrow \infty} \int_a^{a+\Delta y} p_y(\xi) d\xi$$

Approximation
For $y \in \mathcal{Y}$, it holds $p_y(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{y}^n) \forall \xi$

For $x = g(y)$, it holds $p_x(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - g(\bar{y}^n)) \forall \xi$

5.2 Modeling
System *Non-linear, discrete-time:*
$$x(k) = q_{k-1}(x(k-1), v(k-1))$$
$$z(k) = h_k(x(k), w(k))$$

$x(0), \{v\}, \{w\}$ mutually indep. can be D- or CRVs, known PDFs
Objective
Bayesian State Estimator with Monte Carlo Sampling

Init:	$x_m(0)$	$:= x(0)$
S1:	$x_p(k)$	$:= q_{k-1}(x_m(k-1), v(k-1))$
S2:	$z_m(k)$	$:= h_k(x_p(k), w(k))$
	$p_{x_m(k)}(\xi)$	$:= p_{x_p(k)} z_m(k)(\xi \bar{z}(k)) \quad \forall \xi$

Similar proof for $p_{x_p(k)}(\xi), p_{x_m(k)}(\xi) \forall \xi$ as for KF aux. var.

5.3 Particle Filter - Equations
Initialization Draw N samples $\{x_m^n(0)\}$ from $p_x(0)$
S1 – Prior Update / Prediction Step
Apply process equation to the particles $\{x_m^n(k-1)\}$
$$\bar{x}_p^n(k) := q_{k-1}(\bar{x}_m^n(k-1), \bar{v}^n(k-1)), \quad n = 1, \dots, N$$

Requires N noise samples from $p_v(k-1)$

S2 – A Posteriori Update / Measurement Update
Scale each particle by measurement likelihood:
$$\beta_n = \alpha p_{z(k)|x(k)}(\bar{z}(k)|\bar{x}_p^n(k)), \quad n = 1, \dots, N$$

with α normalization s.t $\sum_{n=1}^N \beta_n = 1$

$$\alpha = \left(\sum_{n=1}^N p_{z_m(k)}|x_p(k)(\bar{z}(k)|\bar{x}_p^n(k)) \right)^{-1}$$

Resample to get N posterior particles $\{x_m^n(k)\}$, w/ eq. weights

Repeat N times:
▪ Select random number r uniformly on $(0, 1)$
▪ Pick particle $\bar{n}$ s.t $\sum_{n=1}^{\bar{n}-1} \beta_n < r \leq \sum_{n=1}^{\bar{n}} \beta_n$

New particles all have equal weight
$$p_{x_m(k)}(\xi) \approx \frac{1}{N} \sum_{n=1}^N \delta(\xi - \bar{x}_m^n(k)), \quad \forall \xi$$

Intuition

Process Update – Propagate N particles through system dynamics. Provided N large, $\{x_m^n(\bar{k}-1)\} \approx p_{x_m(k-1)} \rightsquigarrow \{\bar{x}_p^n(k)\} \approx p_{x_p(k)}$
Measurement Update – treat each particle separately. At points of high prior, we have many particles. Posterior has same particles, but scaled by measurement likelihood
5.4 Roughening
Sample Imperovisment *Lumpiness* – when we resample, only retain subset of particles. For finite N this is problem.
Roughening
Perturb particles after resampling

$$\bar{x}_m^n(k) \leftarrow \bar{x}_m^n(k) + \Delta x^n(k)$$

$\Delta x^n(k)$ drawn from 0-mean, finite-var. distribution
Example Method for Δx^n : σ_i stand. dev. of $\Delta x_i^n(k)$

$$\sigma_i = K E_i N^{-\frac{1}{d}}$$

K : tuning parameter, typically $K \ll 1$
 d : dimension of state space
 E_i : max inter-sample variability $\max_{n_1, n_2} |\bar{x}_{m,i}^{n_1} - \bar{x}_{m,i}^{n_2}|$

$N^{-\frac{1}{d}}$: node spacing on grid

6 Observer Based Control
6.1 LTI Observer
LTI System
$$x(k) = Ax(k-1) + Bu(k-1) + v(k-1)$$
$$z(k) = Hx(k) + w(k)$$

 v, w 0-mean noise CRVs
 $\mathbb{E}\{\hat{x}(k)\} \xrightarrow{k \rightarrow \infty} \mathbb{E}\{x(k)\}, \text{Var}\{\hat{x}(k)\}$ bounded
Luenberger Observer
Same structure as steady-state KF:
$$\hat{x}(k) = A\hat{x}(k-1) + Bu(k-1) + K(\bar{z}(k) - \hat{z}(k))$$
$$= (\mathbb{I} - K_\infty H)A\hat{x}(k-1) + (\mathbb{I} - K_\infty H)Bu(k-1) + K_\infty \bar{z}(k)$$
$$\hat{z}(k) = H(A\hat{x}(k-1) + Bu(k-1))$$

Error $e(k) = x(k) - \hat{x}(k)$
in absence of noise $\bar{z}(k) = z(k)$
$$e(k) = (\mathbb{I} - KH)Ae(k-1)$$

- $(\mathbb{I} - KH)A$ must be stable (all $|\lambda| < 1$) for $e \xrightarrow{k \rightarrow \infty} 0$
- Such K exists iff (A, H, A) detectable $\Leftrightarrow (A, H)$ detectable

Observer Design
If (A, H) detectable, construct K
Pole Placement: place λ of e -dynamics of observer modes at desired locations
Steady State KF: Yields optimal K (min steady-state MSE) given statistics of v, w . Also need (A, G) stabilizable, $\text{Var}\{v(k)\} = GG^\top$
Alternative Formulation
$$\hat{x}(k+1) = A\hat{x}(k) + Bu(k) + K(\bar{z}(k) - \hat{z}(k))$$
$$\hat{z}(k) = H\hat{x}(k) \rightarrow \text{delayed measurement}$$

Error Dynamics
$$e(k+1) = (A - KH)e(k)$$

6.2 Static State Feedback Control
System Model
LTI, no noise, deterministic, perf. state info $z(k) = x(k)$:

Feedback Law	CL Dynamics
$u(k) = Fx(k) = Fz(k)$	$x(k) = (A + BF)x(k-1)$

- $(A + BF)$ must be stable (all $|\lambda| < 1$)
- Such F exists $\Leftrightarrow (A, B)$ stabilizable

Controller Design
Pole Placement: place (controllable) poles at des. CL loc.
LQR: find F that min. quad. cost ($\bar{Q} = \bar{Q}^\top \geq 0, \bar{R} = \bar{R}^\top \geq 0$)
$$J_{\text{LQR}} = \sum_{k=0}^\infty x^\top(k) \bar{Q} x(k) + u^\top(k) \bar{R} u(k)$$

LQR Design
Assuming (A, B) stabil., (A, G) detect.
$$F = -(B^\top P B + \bar{R})^{-1} B^\top P A$$

DARE
Same as KF but $A \rightarrow A^\top, H \rightarrow B^\top, Q \rightarrow \bar{Q}, R \rightarrow \bar{R}$
$$P = A^\top P A + \bar{Q} - A^\top P B (B^\top P B + \bar{R})^{-1} B^\top P A$$

6.3 Seperation Principle

Stable Observer + Controller $\rightsquigarrow u(k) = F\hat{x}(k)$

Dynamics

$$\begin{bmatrix} x(k) \\ e(k) \end{bmatrix} = \begin{bmatrix} A + BF & -BF \\ 0 & (\mathbb{I} - KH)A \end{bmatrix} \begin{bmatrix} x(k-1) \\ e(k-1) \end{bmatrix}$$

Separation Principle

- $(\mathbb{I} - KH)A$ and $(A + BF)$ stable $\rightsquigarrow$ **Overall system stable**
- Still stable with noise, Generalizable to time-varying case
- Does **NOT** hold for nonlinear systems

Separation Theorem

Optimal strategy for control problem

$$J_{\text{LQG}} = \lim_{N \rightarrow \infty} \mathbb{E} \left\{ \frac{1}{N} \sum_{k=0}^{N-1} \left[x^\top(k) \bar{Q} x(k) + u^\top(k) \bar{R} u(k) \right] \right\}$$

1. Design steady state KF to provide $\hat{x}(k)$ (**no dependency on $\bar{Q}$, $\bar{R}$**)
2. Design optimal state-FB law $u(k) = Fx(k)$ for determ. LQR problem minimizes J_{LQR} w/o noise (**no dependency on noise statistics Q, R**)
3. Put together